



Whose assets? Individual and household income and savings and mental health in a longitudinal cohort

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ABSTRACT

While there is a growing understanding of the link between financial assets and mental health, the literature is lacking in two dimensions. First, although wealth (including savings) may better inform ability to cope with life's stressors than income alone, the role of wealth in shaping health is substantially understudied relative to income. Second, most studies measure income at the household level instead of the individual level. Using five waves of the nationally representative CLIMB survey collected in Spring 2020, 2021, 2022, 2023, and 2024, we estimated the odds of probable depression (Patient Health Questionnaire, PHQ-9 $\geq$ 10) and anxiety (GAD-7 $\geq$ 10) across individual and household level income and savings using generalized estimating equations to account for clustering over time at the individual level and to account for baseline mental health. Our sample included 1,314 participants across the five waves of the CLIMB survey. The average age of participants was 46.5 years [sd: 16.4] in 2020. When controlling for demographic variables, each financial asset was independently associated with mental health. In fully adjusted models where savings and income at the individual and household level were included, for every \$10,000 more that an adult reported in individual annual income, they had 0.95 times the odds of probable depression (95%CI: 0.92, 0.99) and 0.95 time the odds of probable anxiety (95%CI: 0.91, 0.99), translating to 5% lower odds. Having \$10,000 more in individual savings was associated with 0.95 times the odds of probable anxiety (95%CI: 0.91, 0.99). These findings suggest that 1) financial assets were each independently associated with mental health, 2) having higher individual income was associated with mental health even when adjusting for financial assets at the household and individual level, and 3) higher individual savings were associated with lower probable anxiety. Individual-level financial assets may be associated with mental health differently than household-level financial assets.

1. Introduction

Rates of depression and anxiety have grown for the last 15 years (Goodwin et al., 2020, 2022), with a marked increase during the COVID-19 pandemic (Alqahtani et al., 2023). Globally, over 300 million people live with anxiety and over 280 million people live with depression (World Health Organization, 2022). Younger generations are also reporting higher levels of poor mental health relative to older generations (Santomauro et al., 2021), suggesting that more people may be living with depression and anxiety for a longer time in the coming decades. While poor mental health is common across all demographic groups, there are differences across economic strata (Office of the

Surgeon General (US) et al., 2001; U.S. Department of Health and Human Services, 1999). More than 21 million U.S. adults experienced at least one depressive episode in the previous year (National Institute of Mental Health, 2023) with health care costs and lost productivity estimated to exceed \$326 billion (Cicek et al., 2022; Greenberg et al., 2021). Also in the U.S., between 6% and 44% of adults report elevated symptoms of anxiety (Kavelaars et al., 2023; Terlizzi and Norris, 2021), with treatment costs exceeding \$33 billion (Shirmeshan et al., 2013). Within the U.S., persons with fewer assets are more likely to experience poor mental health, thereby widening health gaps between economic groups (Ettman et al., 2023b). It is therefore of concern that data suggest that the prevalence of depression and anxiety in the U.S. have increased over

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time (Goodwin et al., 2020, 2022), that costs associated with treating poor mental health in the U.S. have grown over time (Greenberg et al., 2021), and that economic disparities persist in who experiences good versus poor mental health over the life course (Macintyre et al., 2018).

Depression and anxiety are responsive to the world around us; exposure to stressors such as traumatic experiences (Kofman et al., 2024), job loss (Pearlin et al., 1981), financial strain (Ettman et al., 2023a,b,c), relationship challenges (Ettman et al., 2022b), or persistent discrimination (Williams et al., 2019) as examples can initiate or exacerbate symptoms of mental illness. Conversely, factors that protect mental health include stable employment, housing, food, transportation, education, and income (Compton and Shim, 2015) as well as the politics and policies that distribute these resources fairly across populations (Dawes and Gonzalez, 2023). In this way, mental health is responsive to multiple levels of influence ranging from the macro-level to micro-environments (Galea, 2017; Kaplan, 2004; Krieger, 1994).

The relationship between wealth and mental health varies across and within national contexts. For example, some higher income countries have a higher prevalence of depressive disorders than some lower income countries (GBD 2019 Mental Disorders Collaborators, 2022). This phenomenon may in part be a function of fewer studies overall in low-income countries, stigma contributing to lower reporting of symptoms, and the potential for screening tools that are not culturally valid in low-income settings leading to under-reporting. However, within countries, there is consistent evidence that higher income populations report better mental health than lower income populations, suggesting some protection of financial assets when controlling for national or cultural context (Patel et al., 2018).

A particularly strong determinant of mental health is higher economic status or access to more assets (Macintyre et al., 2018). In our previous work, we have defined three types of assets—financial, physical, and social—that help shape mental health (Ettman et al., 2023b). Financial assets refer to liquid financial resources such as income, savings, debt, or credit. Physical assets are non-liquid assets that hold monetary value such as cars, homes, and devices that can be used to access other resources (Ettman and Galea, 2024). Finally, social assets refer to assets that bestow privileged treatment through access to additional benefits (such as through marriage) or employment opportunities (such as through education) as examples. This paper is particularly focused on financial assets, which are more liquid, and which can more quickly be used to buffer against stressors that generate poor mental health. Given different societal factors that make access to assets more or less relevant across country contexts, this paper focuses mainly on the U.S., although we draw on literature from other countries where work in the U.S. has not been done to inform our collective understanding of how financial assets may shape mental health.

There is a long-standing body of U.S. literature on the economic gradient that accompanies mental health. Lower socioeconomic status, and particularly lower income, has long been associated with higher rates of mental illness (Dohrenwend, 1966; Kessler and Cleary, 1980; Langner and Michael, 1963; Lorant et al., 2003; Sareen et al., 2011). While causal pathways from income to mental health are challenging to ascertain given inability to randomly allocate people to economic life conditions, there is growing evidence that increases in income improve mental health over time (Kawachi et al., 2010). In a quasi-experiment exploring \$9,000 cash transfers provided to American Indian families from casino revenues in the Eastern Band of Cherokee Indians community as part of The Great Smoky Mountains Study, Costello et al. found that American Indian people (exposed to the cash transfer) had significantly lower prevalence of psychiatric disorders than non-American Indian people who lived in the reservation but did not receive the cash transfers. The research team reported these findings over time: they did not find changes in depression and anxiety symptoms when the adolescents were 15–21 years old (Costello et al., 2003), but in follow up studies, as young adults (through age 21), fewer American Indian participants than non-American Indian participants reported a psychiatric

disorder and the effect of family income supplements were greatest for those who were younger when the income supplements started (and whose family received them over a longer period) (Costello et al., 2010); finally, at the ages of 25 and 30, American Indian people whose families received the income supplements had fewer depression symptoms and anxiety symptoms in adulthood than participants whose families did not receive the income supplements (Copeland et al., 2022).

While there is mounting evidence of the positive and largely monotonic relationship between higher household income and improved mental health, the literature is lacking in two key dimensions. First, much of the literature on economic indicators and mental health excludes non-income financial factors, such as savings or financial strain, which may each relate to mental health differently than income alone. Income captures part, but not all, of a person's financial resources that can serve as a buffer against stressors that in turn provoke or exacerbate depression or anxiety (Benzeval et al., 2014). Income also does not capture the full inequities between groups. For example, in 2016, the median household income for White families was \$61,200 and the median household income for Black families was \$35,400. In comparison, in 2016 the median net worth of White families was \$171,000 and the median net worth of Black families was \$17,600 (Dettling et al., 2017). In 2019, the per-capita income ratio between White and Black persons was 2 to 1 while the per-capita wealth ratio between White and Black persons was 6 to 1 (Derenoncourt et al., 2022).

From 1980 to 2020, while income gaps between Black and White families improved, wealth gaps widened, due in part to capital gains on assets (Derenoncourt et al., 2022). Wealth and savings reflect a person or a family's ability both to earn financial resources and to keep them and accrue them over time. In addition, a system of racist laws and practices in the U.S., such as redlining, Jim Crow laws, discriminatory employment, and unequal access to Veteran's benefits, led to the unequal distribution of wealth among White relative to Black families (Derenoncourt et al., 2022). Therefore, wealth can better capture than income intergenerational advantages or disadvantages faced by populations, and can inform policy solutions to improve health equity and population health moving forward (Fashaw-Walters and McGuire, 2023).

Despite this growing awareness of wealth's association with mental health, income is vastly overrepresented in the literature as an asset related to mental health relative to other assets. In a systematic review of articles on assets and depression during the pandemic, 34 out of 41 of articles featured the relation of income and depression while only 4 out of 41 articles featured savings or related constructs (Ettman et al., 2023a). Thus, there is a need to understand better the relation of non-income financial assets with mental health both alone and when controlling for income.

A second limitation of the current literature on how financial assets shape mental health is ownership and access to the assets that shape health. Most literature on income and mental health uses a household-level indicator for income instead of an individual-level indicator. While it is more common in the survey literature to ask participants to report household income than individual income, theories on access to assets suggest that who has autonomy and access to said resources plays a role in shaping psychiatric outcomes. Thus, understanding whether resources are shared or individually owned may matter for mental health. There are five theories that help to explain how money relates to mental health and how axes of power matter: theories on the stress process, reserve capacity, economic locus of control, unitary household model and resource allocation, and identity capital. The stress process model suggests that debt and economic hardship can deplete mental health both directly by instigating feelings of anxiety and depression and indirectly by eroding a sense of control in one's life (Pearlin et al., 1981). Conversely, access to more resources may buffer individuals' mental well-being from stress-inducing events (Drentea and Reynolds, 2015; Frankham et al., 2020a). Moreover, the reserve capacity model posits that individuals with low socioeconomic status may possess a smaller

“bank” of tangible and intangible resources to buffer stressful events (Gallo et al., 2009; Gallo and Matthews, 2003). The model theorizes that individuals with lower socioeconomic status are exposed to stressful events more often than those with higher socioeconomic status and that the environments individuals with low socioeconomic status disproportionately live and work in stymie the development and replenishment of resources in their reserve (Gallo et al., 2009; Gallo and Matthews, 2003). While the reserve capacity model suggests that the more resources an individual has, the greater their reserve capacity, the economic locus of control theory posits that these resources are more impactful if an individual has access to and control over them (Frankham et al., 2020b; Furnham, 1986). Studies that assess only household-level resources tend to adopt the unitary household model, which assumes that individuals in a household have equal access to financial resources such as income and wealth (Bennett, 2013; Lersch et al., 2022). However, this assumption has been dispelled several times, with intra-household access to resources depending on many factors, such as culture, social dynamics, relationship power dynamics, and gender (Bennett, 2013; Frémeaux and Leturcq, 2020; Lersch et al., 2022). Finally, an individual’s perception of their ability to access needed resources makes it possible for individuals to have control over their environment, autonomy, independence, and ability to bounce back from stressors, all of which shape their idea of self (Butterbaugh et al., 2020) and mental health. The identity capital theory suggests that financial independence is critical for self-actualization and identity formation, as one’s ability to leverage resources allows for a person to have control over one’s time, environment, and life experiences (Butterbaugh et al., 2020). Therefore, considering an individual’s access to resources above and beyond household-level access may be a critical component for understanding the link between financial assets and mental health.

This paper aims to address these gaps in the literature 1) by examining the relation between non-income financial assets (i.e., savings) and mental health and 2) by examining both individual and household level financial assets with mental health in a population-representative group of U.S. adults across time.

2. Methods

2.1. Data

Our data came from the COVID-19 Life Stressors Impact on Mental Health and Well-Being Study (CLIMB), which is a nationally representative survey of U.S. adults (aged 18 years and older). The CLIMB study was fielded via the AmeriSpeak panel, a probability-based sample of households randomly drawn from the NORC sampling frame that has roughly 97% coverage of U.S. households. The CLIMB study required participants in the panel to meet two criteria for inclusion in the initial study; that 1) they are English-speaking, and that 2) they had completed a survey within the last 6 months. The study was fielded over five survey waves spanning March 2020–April 2020 (Wave 1), March 2021–April 2021 (Wave 2), March 2022–April 2022 (Wave 3), March 2023–April 2023 (Wave 4), and March 2024–April 2024 (Wave 5).

1,470 adults participated in the first wave of the survey. Of these, 1,183 participated in Wave 2, 1,093 participated in Wave 3, 938 participated in Wave 4, and 913 participated in Wave 5. In this study, we include participants who responded to at least two survey waves ($N = 1,314$). To account for the possibility of differential dropout, we used custom built survey waves to address attrition over time and to align the sample with the U.S. population. Population representativeness was achieved using post-stratification weights that were benchmarked to the Current Population Survey. Missing demographic variables (i.e., region) were imputed from participant data in other waves.

2.2. Mental health indicators

Probable depression was defined as a binary indicator, measured by the Patient Health Questionnaire-9, which is a commonly used screening tool cited by the U.S. Preventive Services Task Force for screening adults for depression risk (US Preventive Services Task Force et al., 2023a). The PHQ-9 is a validated screener for depressive symptoms that ranges from a score of 0–27, where a score of at least 10 is representative of moderate to severe symptoms and is used as a threshold in clinical care settings to flag for further testing, diagnosis, or specialist referral (Kroenke et al., 2001). Generalized anxiety disorder was also defined as a binary indicator, measured by the Generalized Anxiety Disorder-7, highlighted by the U.S. Preventive Services Task Force as a commonly used screening tool for screening for anxiety disorders (US Preventive Services Task Force et al., 2023b). The GAD-7 is a validated measure of symptoms of generalized anxiety disorder, scored on a scale of 0–21. Studies validating the screener against clinical diagnosis suggest an optimal cutoff of 10 (Spitzer et al., 2006). Our mental health covariates of interest were positive screen for clinical thresholds of depression ($\text{PHQ-9} \geq 10$) and of anxiety ($\text{GAD-7} \geq 10$). Given the clinical significance of exceeding thresholds for these screening tools, we used binary definitions as our primary outcome. However, in the Supplemental Materials, we reran all main models using continuous measure of PHQ-9 and GAD-7 scores. Results were similar across binary and continuous operationalizations of mental health indicators. Baseline mental health was recorded before March 2020 defined by self-reported mental health ranging from poor to excellent.

2.3. Covariables

Our key independent variables included measures of annual income and total savings, measured both at the household level and at the individual level. We defined the financial variables as continuous values derived from participant responses to income categories capped at \$100,000 or greater, measured at each time. Categories ranged from \$0 to \$5,000 (transformed to a midpoint of \$2,500) to greater than or equal to \$100,000 for each of household income, individual income, household savings, and individual savings. Financial variables were then scaled by \$10,000 for regression models. Question stems from the CLIMB study for income and savings measurements are listed in Supplemental Table 1. Given the potential for multicollinearity among these financial measures, we estimated the Variance Inflation Factor (VIF) among them. The average VIF was 1.94 (ranging from 1.08 to 5.31), suggesting low to moderate correlation, so we proceeded with all four measures included.

We included several adjustment variables, including categorical measures of age (18–29, 30–44, 45–59, ≥ 60 years), race and ethnicity (non-Hispanic Black, Hispanic, non-Hispanic other, non-Hispanic White), marital status (married or other), education status (less than high school; high school degree or General Educational Development (GED), some college or vocational/associates degree, bachelor’s degree or above), and respondent’s region of residence (Midwest, Northeast, South, or West), indicator measures of respondent sex and unemployment status, and participant pre-pandemic self-reported mental health status. We also included household size—the number of individuals living in the respondent’s household—measured categorically to account for the total number of persons sharing resources in the home.

2.4. Analysis

First, we described all adjustment variables using population-weighted proportions and unweighted frequencies at Wave 1.

Second, we described the population-weighted proportion of individuals exceeding the clinical thresholds for the PHQ-9 or the GAD-7 across each survey wave, by household and individual income and savings, respectively.

Third, population-weighted generalized estimating equations (GEEs) were used to examine the relationship between exceeding the PHQ-9 or GAD-7 clinical thresholds and both household and individual financial assets, clustering at the participant level to account for repeated measures over time and controlling for baseline mental health. In these equations, we included continuous measures of household and individual-level income and savings separately and in the same model; we adjusted for age, race and ethnicity, sex, marital status, education status, region, unemployment status, household size, and baseline pre-pandemic mental health. We selected an exchangeable correlation structure to account for correlation between repeated measurements over time. We present odds ratios and 95% Confidence Intervals (CI) of exceeding clinical thresholds for screening positive for depression and anxiety, respectively.

To test sensitivity to the presence of adjustment variables, we estimated each model without measures of age, race and ethnicity, sex, employment, education, marital status, region, household size, and baseline mental health. Finally, we re-ran our primary analyses treating CLIMB participants' PHQ-9 and GAD-7 scores as continuous measures to assess the sensitivity of our findings to dichotomization. We presented adjusted regression results and predicted probabilities from fully adjusted models across financial assets. Using the main models, predicted PHQ-9 and GAD-7 scores were estimated at intervals of \$10,000 between \$0-\$100,000 for each financial asset while holding all other covariates constant, either at the reference category for categorical variables or at the mean for continuous variables.

Analyses were conducted in Stata Version 18. Hypothesis testing was two-sided at the 0.05 level. The collection of data through the CLIMB study was deemed exempt by the Institutional Review Board (IRB) at NORC at the University of Chicago. Secondary analysis of de-identified data was deemed not human subjects research by the IRBs at Johns Hopkins School of Public Health and Boston University Medical Center. Informed consent was obtained from all subjects to participate in the study.

3. Results

Table 1 describes characteristics of the analytic sample. In 2020, 29.8% of the weighted sample was aged 60 years or older, 51.7% was female, 62.8% was White, non-Hispanic, 46.8% was married, 34.3% had a bachelor's degree or above, 8.9% was unemployed, and 38.0% lived in the South. **Supplemental Table 2** shows the summary statistics for financial assets at each survey wave.

Fig. 1 shows the weighted prevalence of adults who exceeded the PHQ-9 or GAD-7 clinical thresholds, respectively, by individual and household income and savings levels across each survey wave, from 2020 through 2024. **Supplemental Table 3** and **Supplemental Table 4** show the survey-weighted proportion of U.S. adults with probable depression and probable anxiety over time, respectively. A survey-weighted 39% of adults at Wave 1 with the lowest level of individual savings (\$2,500) had probable depression, while only 8% of adults at Wave 1 with the highest level of individual savings ($\geq$ \$100,000) had probable depression. Overall, adults with greater financial assets had a lower prevalence of probable depression or anxiety at each time point.

Fig. 2 shows the results of models in which household and individual level income and savings were estimated in four separate models per outcome, each including all adjustment variables (demographics and baseline mental health). When modeled separately without other financial assets, higher household income, individual income, household savings, and individual savings were each significantly associated with lower odds of probable depression and probable anxiety, respectively. **Supplemental Table 5** and **Supplemental Table 6** show each individual financial asset model estimating the odds of exceeding clinical thresholds of PHQ-9 and GAD-7, respectively. Higher household income was associated with 0.91 times the odds of probable depression (95%CI: 0.88, 0.94) and 0.96 times the odds of probable anxiety (95%CI: 0.93,

Table 1

Sample demographics: unweighted frequencies and weighted proportions at Wave 1 (2020).

	Unweighted N (Weighted %)
<i>Age</i>	
18–29	198 (20.5%)
30–44	465 (24.7%)
45–59	317 (25.0%)
60+	334 (29.8%)
<i>Race and Ethnicity</i>	
Black, non-Hispanic	121 (11.9%)
Hispanic	226 (16.7%)
Other, non-Hispanic	101 (8.6%)
White, non-Hispanic	866 (62.8%)
<i>Sex</i>	
Male	668 (48.3%)
Female	646 (51.7%)
<i>Marital Status</i>	
Not married	657 (53.2%)
Married	657 (46.8%)
<i>Education</i>	
Less than high school	58 (9.8%)
High school degree or GED	238 (28.2%)
Some college or associate's/vocational	587 (27.7%)
Bachelor's degree or above	431 (34.3%)
<i>Unemployment Status</i>	
Employed	1193 (91.1%)
Unemployed	118 (8.9%)
<i>Region</i>	
Northeast	184 (17.4%)
Midwest	358 (20.7%)
South	439 (38.0%)
West	333 (23.8%)
<i>Baseline Mental Health</i>	
Excellent	278 (21.9%)
Very good	445 (33.4%)
Good	345 (25.8%)
Fair	182 (14.4%)
Poor	57 (4.5%)

Note: Unweighted counts and weighted proportions reflect demographic characteristics collected during Wave 1 of the CLIMB survey in 2020. N = 1314.

0.99), individual income was associated with 0.91 times the odds of probable depression (95%CI: 0.88, 0.94) and 0.93 times the odds of probable anxiety (95%CI: 0.90, 0.97), household savings was associated with 0.95 times the odds of probable depression (95% CI: 0.93, 0.98) and 0.97 times the odds of probable anxiety (95%CI: 0.94, 0.99), and individual savings was associated with 0.94 times the odds of probable depression (95%CI: 0.92, 0.97) and 0.94 times the odds of probable anxiety (95%CI: 0.91, 0.97).

Fig. 3 shows the regression-adjusted odds of exceeding the clinical threshold of the PHQ-9 or GAD-7 associated with changes of \$10,000 of individual and household income and savings levels, respectively. Having \$10,000 more in individual income was associated with 0.95 times the odds of probable depression (95%CI: 0.92, 0.99) and 0.95 times the odds of probable anxiety (95%CI: 0.91, 0.99), even when controlling for household income and any level of savings. Having \$10,000 more in individual savings was associated with 0.95 times the odds of probable anxiety (95%CI: 0.91, 0.99), when controlling for any level of income and household savings.

Table 2 shows the regression-adjusted odds portrayed in **Fig. 1** with coefficients for all covariables included. Between 2020 and 2024, having \$10,000 more in individual savings was associated with 0.95 times the odds (95%CI: 0.92, 0.99) of probable depression and 0.95 times the odds (95%CI: 0.91, 0.99) of probable anxiety, controlling for other financial assets, demographics, and baseline mental health. This translates to 5% lower odds of probable depression or anxiety for every additional \$10,000 in individual income. In that same time period, having an additional \$10,000 in individual savings was associated with 0.95 lower odds of probable anxiety (95%CI: 0.91, 0.99) when controlling for other financial assets, demographics, and baseline mental health. Household

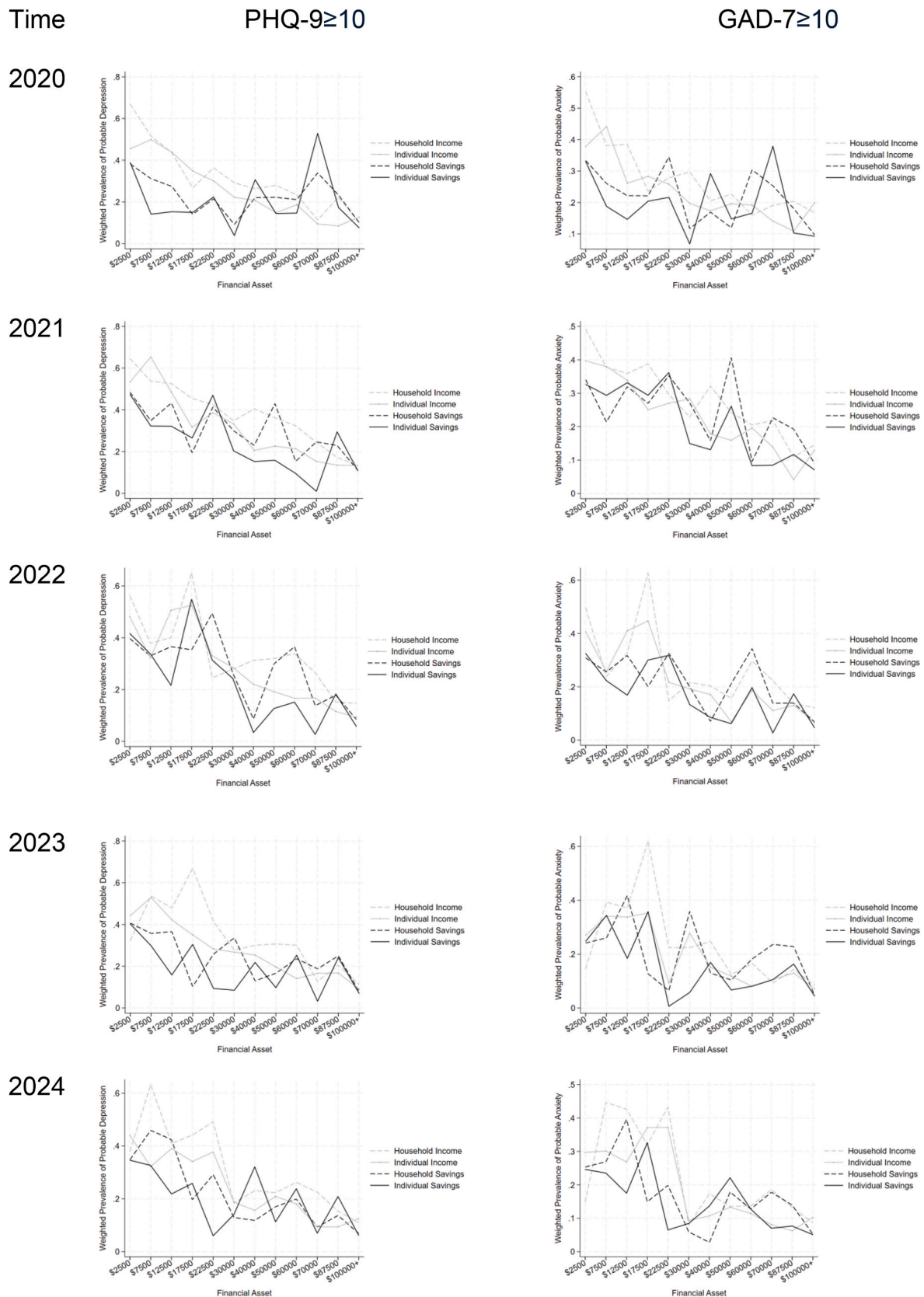


Fig. 1. Weighted prevalence of adults with positive screen for depression (a) or anxiety (b) by household and individual income and savings across time. Note: Weighted proportions describe the sample of individuals that responded to the first (2020) and at least one other survey wave (2021–2024). Dashed lined represents household level financial assets; solid lines represent individual level financial assets. Gray coloring represents income; black coloring represents savings.

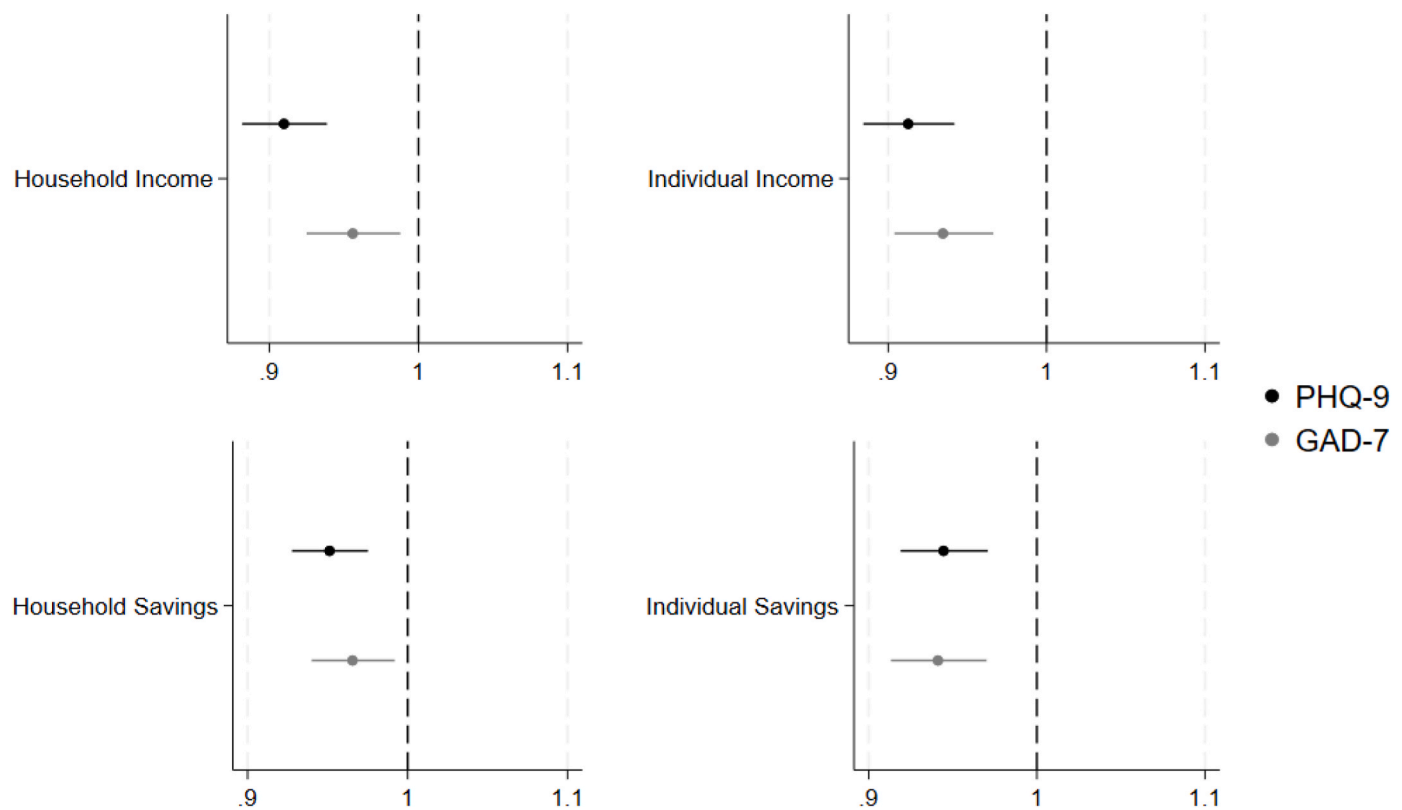


Fig. 2. Odds of exceeding the clinical threshold of the PHQ-9 or GAD-7 adjusted for demographic indicators and baseline mental health: separate financial asset models.

Odds ratios (OR) and 95% Confidence Intervals (CIs) were estimated using generalized estimating equations with an exchangeable correlation structure to account for repeated measures over time. “PHQ-9” indicates being at or above its clinical threshold of 10 out of a possible 27, reflective of having at least moderate depressive symptoms. Similarly, “GAD-7” indicates being at or above its clinical threshold of 10 out of a possible 21, reflective of having at least moderate symptoms of generalized anxiety disorder. Models included: Age, Race and Ethnicity, Sex, Marital and Employment Status, Educational Attainment, Region, Household Size, and Baseline Mental Health.

income but not household savings also had significant associations with probable depression and anxiety.

Supplemental Fig. 1 shows the results of models in which additional covariates beyond household and individual financial assets were not included in the specification. The results of these models were consistent with our primary analyses. Individual income was associated with 0.92 times the odds of probable depression (95%CI: 0.89, 0.95) and 0.93 times the odds of probable anxiety (95%CI: 0.89, 0.96). Individual savings were associated with 0.94 times the odds of probable depression (95%CI: 0.91, 0.97) and 0.92 times the odds of probable anxiety (95%CI: 0.88, 0.95). Supplemental Table 7 shows the results of these models without additional covariates for both PHQ-9 $\geq$ 10 and GAD-7 $\geq$ 10.

Supplemental Fig. 2 shows the results of models in which PHQ-9 and GAD-7 scores were treated as continuous variables (from 0 to 27 for PHQ-9 and 0 to 21 for GAD-7). These findings were mainly consistent with our primary analyses. Supplemental Table 8 shows model coefficients when treating both PHQ-9 and GAD-7 as continuous outcomes. An increase of \$10,000 in individual income was associated with a 0.07 point lower score on the PHQ-9 (95%CI: -0.14, 0.06) and was associated with a 0.07 point lower score on GAD-7 (95%CI: -0.13, -0.003). An increase of \$10,000 in individual savings was associated with 0.09 point lower score on the GAD-7 scale (95%CI: -0.15, -0.02).

Supplemental Fig. 3 shows the predicted probabilities of depressive symptoms (continuous) over time by financial assets adjusting for demographics, time, and baseline mental health, unweighted. Supplemental Fig. 4 shows the predicted probability of anxiety symptoms (continuous) over time by financial assets adjusting for demographics, time, and baseline mental health, unweighted. Findings show that higher financial assets were associated with lower PHQ-9 or GAD-7

scores.

4. Discussion

This paper aimed to address two gaps in the literature on financial assets and mental health. We aimed to incorporate non-income financial assets (savings) and to identify whether who owned the assets (defined by individual versus household income and savings) was relevant to understanding the relationship between financial assets and mental health. First, we found that higher individual income and individual savings were each independently associated with mental health over time, even when controlling for demographics and baseline mental health. Second, in models that accounted for household and individual income and savings together, while adjusting for demographics and baseline mental health, higher individual income was consistently associated with lower odds of reporting clinically meaningful depression or anxiety. Third, having higher individual savings was associated with lower odds of probable anxiety over time, when controlling for other assets, demographics, and baseline mental health. These findings suggest that individual-level financial assets may be associated with mental health differently than household-level financial assets. Who has access to financial resources—and who contributes to financial assets—in a household may be contributory to mental health.

This paper contributes to a growing literature exploring how financial assets beyond income, such as wealth and savings, relate to mental health. Despite there being fewer studies on wealth and health relative to income and health (Ettman et al., 2023b; Pollack et al., 2007), there is growing consensus that wealth and other financial assets should be considered as formative factors in understanding what shapes health

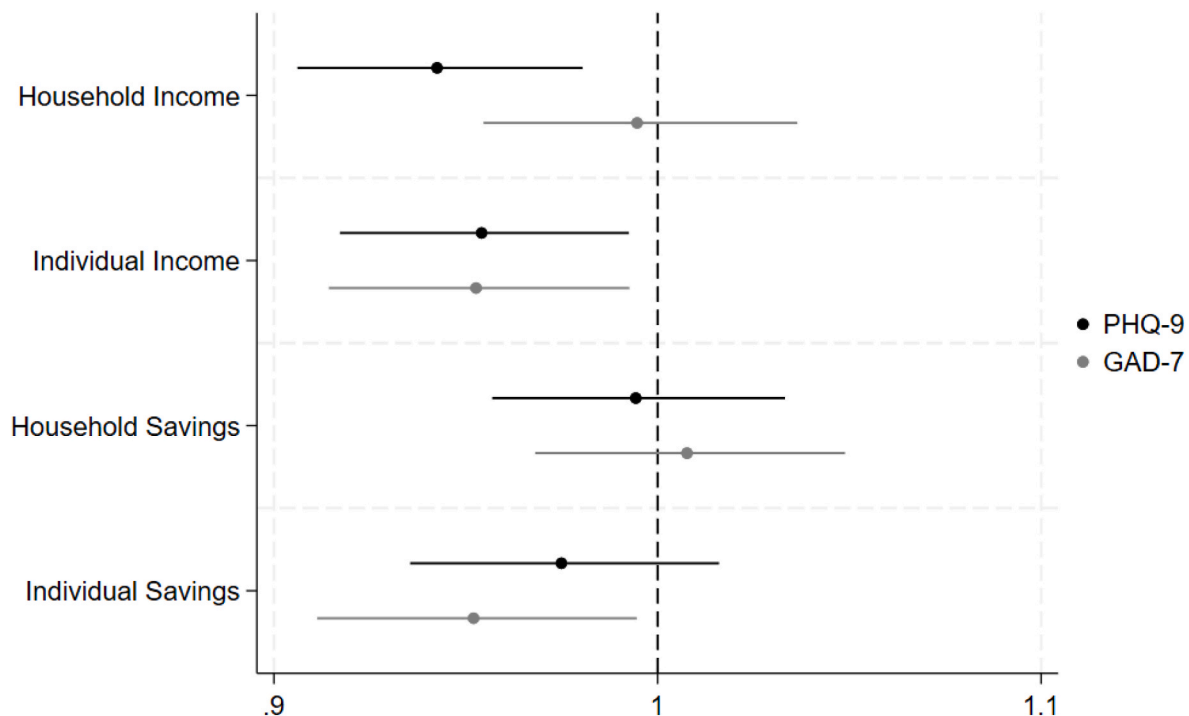


Fig. 3. Odds ratios of exceeding the clinical threshold of the PHQ-9 or GAD-7 over time by household and individual income and savings together adjusted for demographics, region, and baseline mental health.

Note: Odds ratios and 95% CIs were estimated using generalized estimating equations with an exchangeable correlation structure to account for repeated measures over time. “PHQ-9” indicates being at or above its clinical threshold of 10 out of a possible 27, reflective of having at least moderate depressive symptoms. Similarly, “GAD-7” indicates being at or above its clinical threshold of 10 out of a possible 21, reflective of having at least moderate symptoms of generalized anxiety disorder. Models included: Household income, Household savings, Individual income, Individual savings, Age, Race and Ethnicity, Sex, Marital and Employment Status, Educational Attainment, Region, Household Size, and Baseline Mental Health.

(Dean and Nicholas, 2018; Ettman and Galea, 2024). Our findings are aligned with studies in other countries exploring financial factors beyond income on mental health, which consistently find that lower financial assets are related with elevated symptoms of mental illness (Bialowolski et al., 2024; Parra-Mujica et al., 2023; Kourouklis et al., 2020; Melchior et al., 2010; Oskrochi et al., 2018). Our study found that having higher individual savings was associated with lower odds of positive screen for anxiety even when controlling for income and other demographic characteristics. A household panel study in Australia compared the importance of income versus wealth on well-being through comparison of different coefficient sizes and through estimating differences in well-being around different distributions of income and wealth. They found that wealth had a greater influence on well-being than income when comparing sharply different distributions of income and wealth (Headey and Wooden, 2004). In a representative sample of middle-aged and older Israeli and European adults, Bialowolski et al. also compared the relation of different types of financial assets with mental health. They found that the protective nature of financial assets on mental well-being may differ based on the liquidity of the asset, with more liquid assets, such as funds in a bank account, generally being more influential on mental health than less liquid assets, such as retirement savings (Bialowolski et al., 2024). In another study that compared income with non-income financial indicators in a nationally representative study of British heads of households, perception and expectations of future financial well-being and problems meeting household needs were significantly associated with depression while household income was not (Oskrochi et al., 2018), adding evidence to the role of non-income financial factors (including perception of ability to cover expenses) in shaping mental health. In a study of older adults across Europe, Kourouklis et al. found a more consistent association between wealth and mental health than income and mental health. However, they found that the association was weaker

in Nordic countries, which have a strong social safety net (Kourouklis et al., 2020), which suggests that policy contexts may influence the strength of the relation between financial assets with mental health. Within the U.S., in a study of 18–27 year-olds, having household wealth in the top quartiles compared to the lowest quartile in childhood was associated with a statistically lower prevalence of mental illness in young adulthood in models that did not adjust for household income (Lê-Scherban et al., 2016); the association between childhood wealth and later psychological distress was attenuated by household income and parental education. Their finding that the association between wealth and mental health was attenuated by household income is consistent with our study findings that showed that both household savings and individual savings were significantly associated with mental health when considered in separate models but whose associations were attenuated when adjusting for household and individual income.

While much of the literature on financial assets measures household-level finances, a paucity of literature exists documenting the link between individual-level financial assets and mental health. A nationally representative household panel study of Polish adults found that people with individual savings reported significantly higher emotional health (but no significant association with PHQ score) than those without savings (Bialowolski et al., 2019). In a study assessing individual income in a longitudinal, national registry of over 1.2 million Finnish adults, individuals in the lowest income quintile at age 30 incurred a higher risk of being diagnosed with a mental disorder at the follow-up period compared to individuals in the highest income quintile, even after accounting for prior history of mental disorder (Hakulinen et al., 2023). These findings are consistent with a 2011 meta-analysis showing a positive relationship between individual autonomy and mental well-being and between financial assets and individual autonomy (Fischer and Boer, 2011). Similarly, a 2020 study revealed that individual agency plays an important role in the association between

Table 2

Association between household and individual income and savings and the odds of exceeding the clinical threshold of the PHQ-9 or GAD-7 over time, adjusted for demographics and baseline mental health.

Characteristics	PHQ-9 ≥ 10				GAD-7 ≥ 10			
	OR	95% CI		P-value	OR	95% CI		P-value
Household income	0.94	0.91	0.98	0.003	0.99	0.95	1.04	0.800
Individual income	0.95	0.92	0.99	0.020	0.95	0.91	0.99	0.021
Household savings	0.99	0.96	1.03	0.772	1.01	0.97	1.05	0.708
Individual savings	0.97	0.94	1.02	0.228	0.95	0.91	0.99	0.028
Age								
18–29	2.06	1.50	2.83	<0.001	2.49	1.77	3.50	<0.001
30–44	2.07	1.56	2.74	<0.001	2.67	1.97	3.61	<0.001
45–59	1.38	1.07	1.79	0.014	1.57	1.18	2.08	0.002
≥ 60	Ref	Ref	Ref		Ref	Ref	Ref	
Race and ethnicity								
White, non-Hispanic	Ref	Ref	Ref		Ref	Ref	Ref	
Black, non-Hispanic	0.97	0.68	1.37	0.856	0.87	0.60	1.27	0.471
Hispanic	0.99	0.73	1.35	0.969	1.00	0.72	1.37	0.984
Other, non-Hispanic	1.13	0.77	1.65	0.527	0.94	0.63	1.42	0.786
Sex								
Male	Ref	Ref	Ref		Ref	Ref	Ref	
Female	1.22	0.98	1.51	0.075	1.23	0.98	1.54	0.079
Marital status								
Not married	Ref	Ref	Ref		Ref	Ref	Ref	
Married	0.81	0.66	1.01	0.057	0.93	0.74	1.16	0.522
Employment								
Not employed	Ref	Ref	Ref		Ref	Ref	Ref	
Employed or other	0.61	0.48	0.79	<0.001	0.72	0.56	0.92	0.010
Education								
Less than high school degree	0.79	0.53	1.18	0.251	1.03	0.67	1.58	0.895
High school degree or GED	1.39	1.05	1.83	0.022	1.34	0.99	1.81	0.056
Some college or vocational/technical school	1.32	1.02	1.72	0.038	1.42	1.07	1.88	0.015
Bachelor's degree or above	Ref	Ref	Ref		Ref	Ref	Ref	
Region								
Northeast	Ref	Ref	Ref		Ref	Ref	Ref	
Midwest	0.70	0.50	0.98	0.041	0.70	0.49	1.00	0.047
South	0.75	0.56	1.01	0.061	0.78	0.57	1.07	0.122
West	0.78	0.56	1.09	0.141	0.61	0.43	0.87	0.007
Household size								
1	Ref	Ref	Ref		Ref	Ref	Ref	
2	1.14	0.90	1.44	0.271	0.81	0.63	1.03	0.086
3	1.06	0.81	1.39	0.686	0.83	0.63	1.10	0.193
4	1.42	1.07	1.88	0.014	1.03	0.77	1.37	0.866
5	0.97	0.71	1.34	0.868	0.49	0.35	0.70	0.000
6 or more	0.89	0.64	1.23	0.486	0.71	0.51	0.99	0.042
Baseline mental health								
Excellent	Ref	Ref	Ref		Ref	Ref	Ref	
Very good	1.24	0.89	1.74	0.208	1.16	0.80	1.67	0.441
Good	2.13	1.53	2.97	<0.001	1.84	1.28	2.64	0.001
Fair	6.38	4.42	9.19	<0.001	4.29	2.92	6.29	<0.001
Poor	15.13	8.53	26.82	<0.001	15.73	8.96	27.64	<0.001
_cons	0.33	0.19	0.57	0	0.19	0.11	0.34	0

Note: Odds ratios (OR) and 95% Confidence Intervals (CIs) were estimated using generalized estimating equations with an exchangeable correlation structure to account for repeated measures over time and baseline mental health. “PHQ-9” indicates being at or above its clinical threshold of 10 out of a possible 27, reflective of having at least moderate depressive symptoms. Similarly, “GAD-7” indicates being at or above its clinical threshold of 10 out of a possible 21, reflective of having at least moderate symptoms of generalized anxiety disorder.

financial assets and mental well-being (Frankham et al., 2020a). Finally, our study contributes evidence against the unitary household model, which suggests that monetary resources are shared equally among individuals in a household. Other recent studies reveal high levels of intra-household inequality (Lersch et al., 2022). The unitary household model may particularly overestimate the amount of wealth held by women in heterosexual married couples (Bennett, 2013; Frémeaux and Leturcq, 2020). Thus, incorporating individual-level economic exposures over household-level economic exposures when possible may improve research and inform the relation between financial assets and formation of mental health.

Policy implications of this work depend, in part, on causal assumptions of the relation between financial assets and mental health. While promoting economic facilities is one of several key pillars to human development (Sen, 2011), the direction of causality between economic improvement and mental health is an open area of inquiry and it is likely

that the relationship is bidirectional (Arcaya et al., 2015; Kawachi et al., 2010). While there is some evidence that being diagnosed with depression or anxiety in early adulthood is linked with lower employment and earnings in later adulthood (Hakulinen et al., 2019), there is also evidence that positive income shocks improve mental health (particularly for women) (Golberstein, 2015) and that negative wealth shocks lead to increased depressive symptoms (Pool et al., 2017). In a study of adults in France and the UK from September 2022–August 2023, Nettle et al. used exogenous shocks of price increases (which reduce the ability to save) to study the effect of reductions in financial wealth on mental health; they found that increases in monthly costs of essential goods had an immediate influence on depression (Nettle et al., 2024). Social policy implications of their work suggest that policies should consider smoothing financial shocks such as income changes and price increases, which can reduce savings and affect mental health in short order. Studies comparing objective and subjective financial

standing and mental health suggest that subjective financial standing may be more strongly associated with mental health than objective indicators of wealth (Villadsen et al., 2024; Wilkinson, 2016). Additionally, low wealth could lead to an accumulation of health-reducing exposures (Nettle et al., 2024), such as living in neighborhoods with more pollution or greater violence, which could require a multi-pronged approach to asset building to improve mental health. Therefore, not all efforts to increase objective financial standing may result in immediate mental health gains.

Policy interventions to improve financial standing include unconditional cash transfers such as universal basic income and conditional transfers such as child development accounts or “baby bonds” (Sun et al., 2021). Evidence of health changes has been mixed, although improved mental health is a more consistent finding than improved physical health indicators. In a randomized study of residents in low-income Chelsea, Massachusetts, individuals who received a \$400 cash transfer monthly for 9 months via debit card had fewer emergency department visits for behavioral health than individuals who applied to the cash assistance lottery but did not receive the funds (Agarwal et al., 2024). In the SEED Oklahoma experiment, which randomized mothers to receive a \$10,000 state-owned 529 education savings account for their children and matching incentives for self-owned 529 education savings accounts for their children versus a control group with no incentives, mothers in the treatment group reported reduced depressive symptoms more than three years later (Huang et al., 2014). While more than half of the literature on wealth and depression suggests that wealth is associated with better mental health (Ettman et al., 2022a), the one study to our knowledge that suggests a negative effect of wealth on mental health found that moving to a more affluent neighborhood led to worsened mental health symptoms for adolescents (Nieuwenhuis et al., 2017). Thus, while the current study suggests that policies that improve individual savings may improve mental health, the success of interventions to bolster economic standing within and across households depends on multiple factors, including timing, delivery, frequency, duration, and intended recipients—as well as the contexts in which people live. Finally, social comparison, relative wealth, inequality, and costs-to-assets-ratios may all influence the relation of wealth with mental health, which may influence the mental health effects of policy interventions to improve financial well-being.

This study should be considered in light of several main limitations. First, our main outcome of interest is clinically relevant thresholds for positive screen for depression and anxiety. Both screeners used measure symptoms of depression and anxiety, respectively, and do not replace a clinical diagnosis from a provider. However, the PHQ-9 and the GAD-7 have been shown to have high sensitivity and specificity when compared to the gold standard of diagnosis by a provider (Kroenke et al., 2001; Spitzer et al., 2006). These measures allow for population-level monitoring of mental health, which is harder to achieve via formal diagnosis which embeds bias through uneven access to mental health services. Second, the relation between mental health and financial resources can be bi-directional; that is, while we explored these findings through the lens of how having fewer financial assets may be associated with worse mental health, it is possible that having worse mental health also leads to having fewer financial assets. That this study is longitudinal helps to mitigate some of that concern, suggesting that the relation between financial assets and mental health is sustained over time. Third, this study was conducted in Spring of 2020, 2021, 2022, 2023, and 2024 which overlapped with the COVID-19 pandemic. Despite being collected during the COVID-19 pandemic, these findings may reflect trends that are generalizable outside of the pandemic-era. Use of the pooled longitudinal data yielding consistent trends over time suggests that the differences in mental health across financial assets were sustained outside of the COVID-19 era. It is possible, even, that associations between financial assets and mental health would be even greater in the absence of generous social and economic policy passed during the pandemic that supported low- and middle-income families.

5. Conclusion

This study is one of the first to assess the relation of both income and savings at the individual and household level with mental health. Our finding that higher individual income but not household financial assets was consistently associated with poor mental health suggests that individual ownership of assets may matter for the protection of mental health. Further research may fruitfully assess the bidirectional relation between financial assets and mental health. This research may productively explore non-income financial assets in addition to income and individual ownership in addition to household ownership of assets. Policies that aim to improve mental health through the accrual of assets may consider individuals’ access to those resources to optimize the potential beneficial impact of such policies on population mental health.

CRedit authorship contribution statement

Catherine K. Ettman: Writing – review & editing, Writing – original draft, Project administration, Investigation, Funding acquisition, Conceptualization. **Emma Dewhurst:** Writing – review & editing, Writing – original draft, Resources. **Rajesh Satpathy-Horton:** Formal analysis, Methodology, Visualization, Writing – review & editing. **C. Ross Hatton:** Writing – review & editing, Methodology, Formal analysis. **Ben Thornburg:** Writing – original draft, Investigation, Formal analysis. **Brian C. Castrucci:** Writing – review & editing, Resources, Funding acquisition. **Sandro Galea:** Writing – original draft, Supervision, Resources, Conceptualization.

Declaration of competing interest

The authors have no competing interests to declare. The de Beaumont Foundation funded the CLIMB survey Waves 2–5. BCC and ED work at the de Beaumont Foundation. They were involved in the revision of survey questions (ED and BCC), allocation of resources (BCC), manuscript – writing (ED), and manuscript – editing (ED and BCC).

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.socscimed.2025.117813>.

Data availability

Data will be made available on request.

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